



Cambridge (CIE) IGCSE Computer Science



Your notes

Databases

Contents

- * Single Table Databases
- * Primary Keys



Your notes

Single Table Databases

What is a database?

- A **Database** is an organised collection of data
- A database is made up of either **one or multiple tables** which are made up of **fields** and **records** to organise how it stores data
- It allows **easy storage, retrieval, and management** of information
- A database is useful when working with **large amounts of data**, databases are stored on **secondary storage**
- A database is often stored on remote servers so **multiple users** can access it at the same time, useful for online systems
- Data can be **sorted and searched efficiently**, making use of more advanced structures
- They are **more secure** than text files

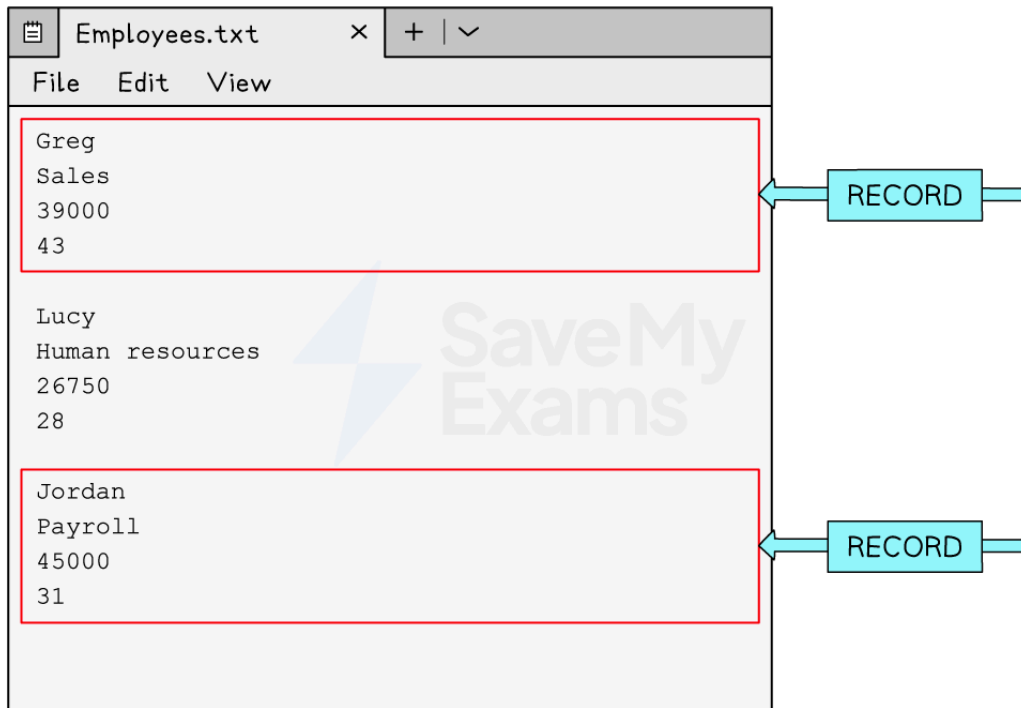
Fields & records

What are fields & records?

- A field is **one piece** of information relating to one person, item or object
- A field is represented in a database by a **column**,
- A record is a **collection of fields** relating to one person, item or object
- A record is represented in a database by a **row**

Text files

- A text file is useful when working with **small amounts of data**, text files are stored on **secondary storage** and 'read' into a program when being used
- They are used to store information **when the application is closed**
- Each entry is stored on a new line or separated with a special identifier, for example, a comma (',')
- It can be difficult in text files to know where a record begins and ends



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Validation

- When a table is created, **validation rules** can be assigned to the different fields
 - A validation rule **controls what data** can be entered into that field
 - There are different types of validation checks used to limit what data can be entered into each field
 - The validation checks that are used are the same as those that appear in section 7: Algorithm Design & Problem-Solving

Type	Description
Length Check	This type of validation checks the number of characters that have been entered into a field. For example, you might make phone numbers entered have to be eleven characters long
Format Check	This type of validation checks data entered meets an exact format. For example, a product code might have to be two letters followed by five numbers



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Range Check	A range check will check the number entered is within a specific range. For example, the age of a dog would be between 0 - 40. Any other number would be disallowed
Presence Check	A presence check can be added to fields which cannot be left blank
Type Check	A type check will allow data with a specific data type to be entered into a field. For example, if text was entered into a field which is supposed to contain the date of birth of a person it would not be allowed
Check Digits	Check digit validation is a process used to verify the accuracy of numbers such as credit card numbers. A check digit is a single digit added to the end of the number, which is calculated based on a specific algorithm applied to the other digits in the number. When the data is re-entered the same algorithm can be applied, and if it produces a different result the code is incorrect



Examiner Tips and Tricks

You will likely be presented with an example database table and identify either how many fields there are or how many records there are so make sure you remember a record is a row and a field is a column.



Worked Example

A Database Table Containing Student Grades

StudentID	FirstName	LastName	MarkSubmitted	Percentage
1483792	Shanay	Giles	Y	55
1498378	Poppy	Petit	N	20
1500121	Diya	Dinesh	Y	74
1382972	Joe	Swaile	Y	68
1598264	Anton	Smith	Y	34



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1548282	Felicity	Hall	N	47
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Describe **two validation checks** that could be used to check data being inputted into the table above [4]

How to answer this question

This

Answer

StudentID could have a **length check** to ensure 7 characters are entered [2]

FirstName and LastName could have a **presence check** to make a record cannot be entered without entering the name of the student [2]

A **type check** of boolean could be applied to the Mark Submitted field so that only Y or N are entered [2]

A **range check** could be assigned to the Mark column to ensure only numbers between 0 and 100 are entered [2]

Data types

What is a data type?

- A data type is the **type of data** that can be held **in a field** and is defined when designing a table
- Examples of common datatypes are:
 - **Integer** - whole number
 - **Real** - decimal number
 - **Text/alphabetic** - text data
 - **Character** - single
 - **Date/Time**
 - **Boolean** - true or false values
- In the table **cars** below, the following datatypes would be used:
 - **car_id**: integer
 - **make**: text/alphabetic
 - **model**: text/alphabetic
 - **colour**: text/alphabetic

▪ price: real

cars

car_id	make	model	colour	price
1	Peugeot	2008	Red	24950
2	Mazda	MX5	Blue	17995
3	Citroen	DS4	Black	21450
4	Ford	Puma	White	19500



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Primary Keys

Primary Keys

What is a primary key?

- A primary key is a **unique field** that can be used to **identify a record** in a **table**
- `customer_id` is the **primary key** for the customers **table** below
- Other examples of primary keys in database tables would include:
 - `Student_ID` in a school database
 - `Car_Registration` in a car database
 - `Product_ID` in a shop database

CustomerID	FirstName	LastName	DOB	PhoneNumber
001	Andrea	Bycroft	05031976	0746762883
002	Melissa	Langler	22012001	0756372892
003	Amy	George	22111988	074637

Key database terminology

Term	Definition
Table	A collection of records with a similar structure
Record	A group of related fields, representing one data entry
Field	A single piece of data in a record
Data type	Type of data held in a field



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Primary key	A unique identifier for each record in a table. Usually an ID number
Foreign key	A field in a table that refers to the primary key in another table. Used to link tables and create relationships



Worked Example

A database has been developed for a dance club to store information about their members.

The database contains one table: **Members**

Figure A shows some data from the table.

Members

MemberID	FirstName	LastName	DateJoined
1	Zarmeen	Hussain	2024-01-19
2	Fyn	Ball	2024-02-01
3	George	Johnson	2024-02-25
4	Ella	Franks	2024-03-04

State the name of the field from the **Members** table that is the most suitable to use as the primary key [1]

Answer

- (a) *MemberID*